B.Tech 6th Semester Mini project Presentation

"Microplastic degradation by Algae for Sustainable Development"

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Content of Presentation

1. Abstract
2. Introduction
3. Problem Statement
4. Literature Review
5. Project Module
6. Use case diagram/ER diagram/Flow diagram
7. Excepted Outcome of the Project
8. Limitation
9. Conclusion and Future Work
10. References

Abstract

- Microplastic pollution has emerged as critical environmental challenge due to its persistence in ecosystem and adverse effects on biodiversity.
- These tiny plastics particles, originating from industrial sources and the breakdown of large plastics, have been detected in waterbodies, soil, and even the atmosphere.
- Conventional removal methods such as filtration ,chemical treatments, and incineration have shown limited efficiency and often contribute to further environmental damage.
- As a sustainable alternative, microalgae based bioremediation offers a promising solution. Certain species, including *Chlorella vulgaris* and *Dunaliella salina*, can degrade microplastics through enzymatic activity while simultaneously supporting biofilm formation on plastic surfaces, accelerating biodegradation.
- Furthermore, algae-derived bioplastics present an eco-friendly alternative to petroleum-based plastics, addressing both plastic waste management and sustainability concerns.
- This review explores the potential of microalgae in tackling microplastic pollution, discussing its mechanisms, challenges, and future applications in large-scale wastewater treatment.

Introduction

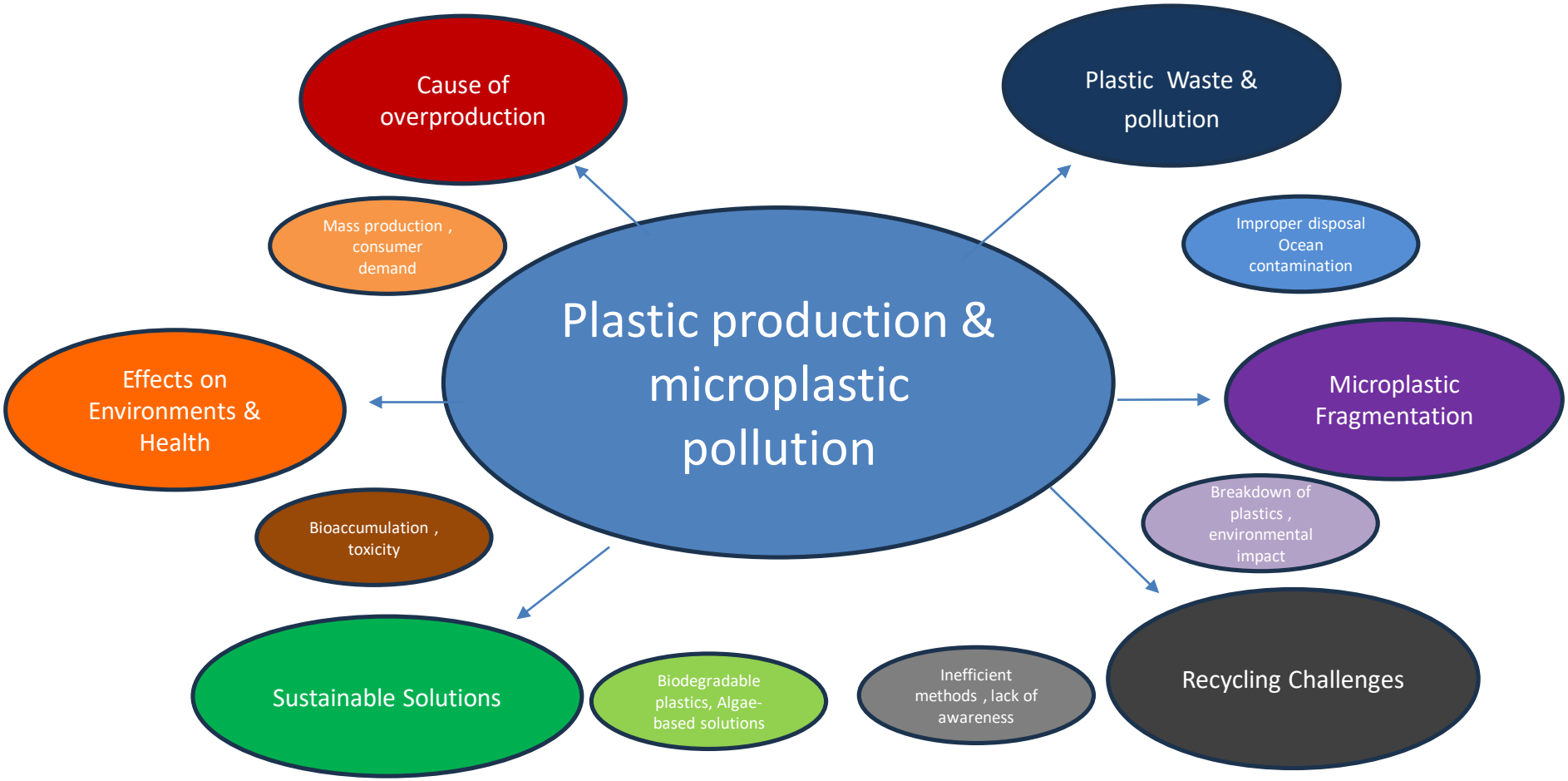
- Plastic pollution has become an **alarming global crisis**, with microplastics defined as plastic particles smaller than 5 mm—posing a significant environmental threat. With global plastic production exceeding **359 million metric tons** in 2018 and projected to triple by 2050, the accumulation of plastic waste has reached **critical levels**. Microplastics originate from both **primary sources** (such as cosmetic microbeads and industrial abrasives) and **secondary sources** (resulting from the breakdown of larger plastic items). These particles infiltrate aquatic ecosystems, soil, and even the air, leading to long-term ecological and health risks.

- Despite various mitigation strategies, conventional plastic removal methods remain **inefficient and unsustainable**. Wastewater treatment plants have shown **low microplastic removal rates (often below 50%)**, while incineration and chemical treatments contribute to **secondary pollution** in the form of toxic emissions. Additionally, recycling rates remain **alarmingly low**, with less than **10% of plastic waste** effectively repurposed. These limitations highlight the urgent need for **sustainable, biological solutions**.

- Microalgae-based bioremediation has gained attention as a **natural, cost-effective** approach to degrading microplastics. Certain algal species can **attach to plastic surfaces**, forming biofilms that facilitate enzymatic breakdown. Moreover, algae-derived bioplastics offer a **renewable alternative** to conventional plastics, reducing dependency on fossil fuels. This review explores the role of **microalgae in microplastic degradation**, analyzing the mechanisms, challenges, and future applications of algal bioremediation in wastewater treatment and environmental conservation.

PROBLEM STATEMENT

Microplastic Pollution: A Growing Crisis & A Sustainable Solution



LITERATURE REVIEW

Title of the paper	Algae speices used	Key points	References
Unveiling the Potential: Biodegradation of Plastics by Algae	Dunaliella salina, Chlamydomonas reinhardtii, Scenedesmus dimorphus	Biodegradation Process, Role of Algae in Plastic Degradation,	Agate, S., Shivalkar, S., Khan, A., & Soni, S. (2024).
Biodegradation of Low-Density Polyethylene (LDPE) Sheet by Microalga, Uronema africanum Borg	Uronema africanum Borge	Hydrocarbons (e.g., Nonane, Tricosane, Tetracosane), Fatty Acids (e.g., n-Hexadecanoic acid)	**Scientific Reports (2021), DOI: 10.1038/s41598-021-96315-6**【17
Nature's Fight Against Plastic Pollution: Algae for Plastic Biodegradation and Bioplastics Production	Chlorella vulgaris, Scenedesmus dimorphus, Anabaena spiroides, Oscillatoria princeps, Phormidium lucidum	Bioplastics (PHA, PHB), Hydrocarbons, Carbohydrates	**Environmental Science and Ecotechnology (2020), DOI: 10.1016/j.ese.2020.100065**【19】

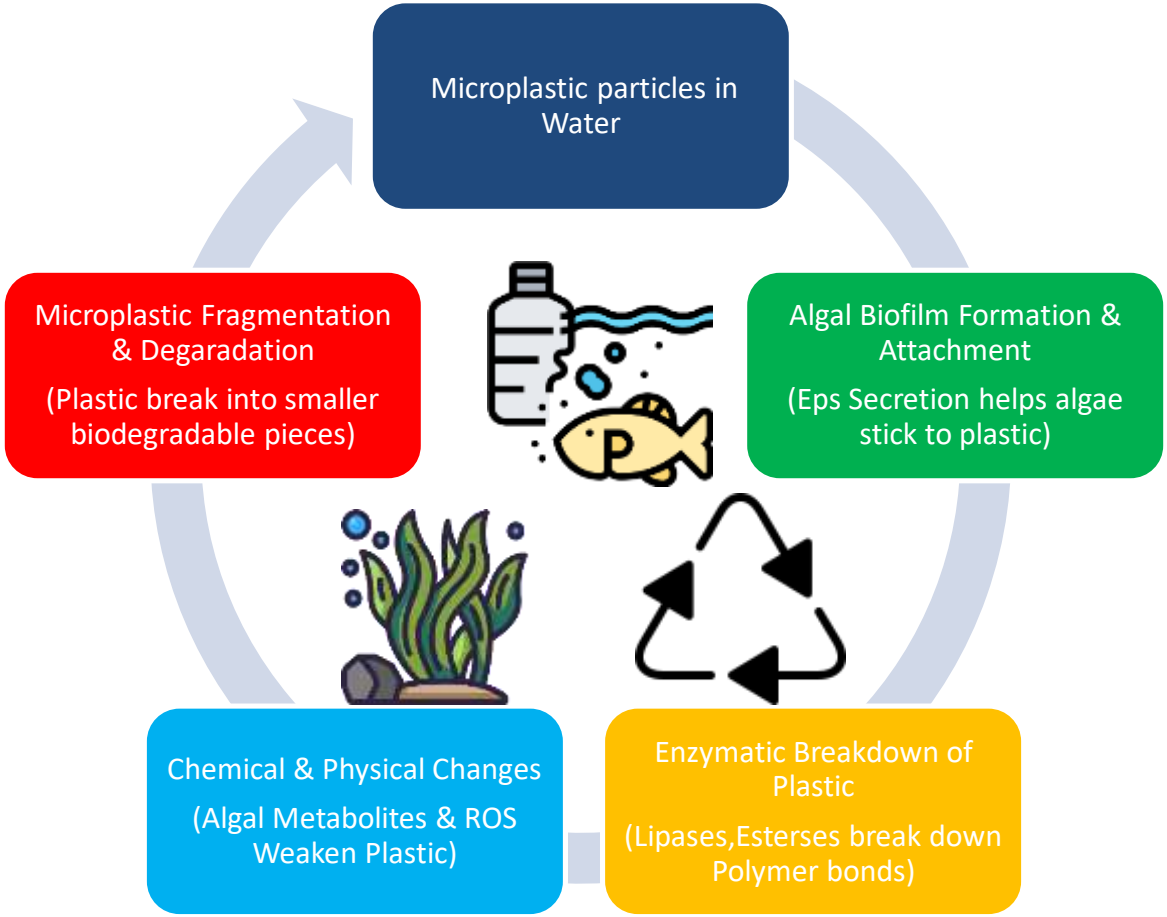
Objectives

1. To screen the potential microalgae for high biomass production
2.To analyse the mixotrophic cultivation of selected microalga
3.To emphasize the degradation of bioplastics by microalgae
.

Expected Outcome

- **High Microplastic Removal Efficiency:**
- Expecting **70–98% degradation** under optimized lab conditions.
- **Improved Water Quality:** Reduction in toxic plastic residues, heavy metals, and pollutants .
- **Scalability Potential:** Feasibility of large-scale wastewater treatment integration .
- **Eco-Friendly Alternatives:** Algae-derived bioplastics as a sustainable replacement for conventional plastics.

FLOW CHART


Microplastic particles in Water
Algal Biofilm Formation & Attachment
(Eps Secretion helps algae stick to plastic)
Enzymatic Breakdown of Plastic
(Lipases,Esterses break down Polymer bonds)
Chemical & Physical Changes
(Algal Metabolites & ROS Weaken Plastic)
Microplastic Fragmentation & Degaradation
(Plastic break into smaller biodegradable pieces)

Case Studies & Experimental Evidence

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Several Studies have explored how microalgae interact with microplastics , influceing degradation , photosynthesis , and aggregation.

Algae species	Microplastic type	Degradation Efficiency (%)	Key Obeservations
Chlorella vulgaris	Polysterene (PS)	73.01%	Optimal at PH 7.5,reaction time 31.90 min
Chlorella Pyrenoidosa	Polyethelene (PE),PS	Negative impact on photosynthesis	50% deop in Fv/Fm ratio,40% reduction in chlorophyll a
Scenedesmus abundans	Mixes MPs	Upto 98% removal	Effective in waste water treatment
Chlorella sp. + Scenedesmus sp.	Domestic wastewater MPs	78-98% Nutrient removal	Combined species enhance plastic breakdown

Advantages of Algae-Based Microplastic Removal

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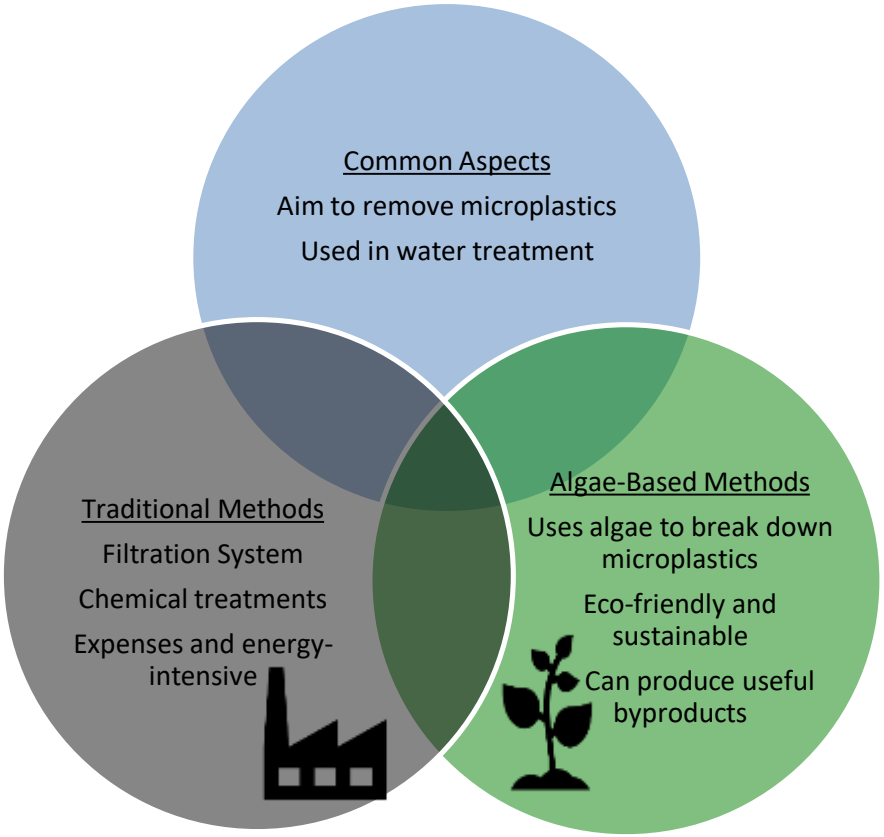
" Comparision of Algae-Based and Traditional Microplastic Removal Methods "

Aspect	Algae-Based Methods	Traditional Methods
Sustainability	Eco-friendly, natural process, minimal byproducts.	Uses chemicals,risk of toxic byproducts.
Cost-Effectiveness	Budget-friendly, low operational costs.	Expensive, require chemical production.
Scalability	Easily integarated into wastewater plants.	Limited scability, high infrastructure costs.
Efficiency	Up to 94.5% removal efficiency.	Varies, sometimes lower efficiency.
Environment Impact	Produces biodegradable waste, reusable as biofertilizer.	Generates toxic sludge, requires disposal.

Comparision of Microplastic Removal methods



Challenges & Limitations

Efficiency Issues: Slow Degradation Rate

- Not all algae strains can break down plastics effectively.
- The process depends on enzymes, which some algae don't produce.
- Certain pollutants, like pharmaceuticals, are tough for algae to remove.

Environmental Factors Affecting Algae Performance

- Temperature, pH, and nutrients all play a role in efficiency.
- Warm temperatures boost algae growth, but extreme heat can be harmful.
- High nitrogen and phosphorus levels speed up growth but can also trigger harmful algal blooms.

Risks of Algal Blooms

- Some algae produce toxins that harm fish, marine life, and even humans.
- Massive blooms deplete oxygen in the water, creating dead zones.
- Climate and pollution make blooms more common, making prevention crucial.

CONCLUSION AND FUTURE WORK

•**Conclusion:**

- Microalgae-based bioremediation is a promising, sustainable approach for microplastic degradation
- Certain algae species demonstrate **high plastic removal efficiency** and contribute to improved water quality .
- The process is **cost-effective and scalable** for wastewater treatment applications.

•**Future Work:**

- **Enhancing Algal Efficiency:** Genetic modifications or co-cultivation strategies to improve degradation .
- **Field Studies:** Large-scale testing in real-world polluted water bodies .
- **Bioproduct Development:** Exploring algae-derived bioplastics for commercial applications.

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